

Rigers Aliaj

Data Engineer, Scientific Data Workflows, AI Features and Optimization

Hamburg, Germany

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Profile

PhD researcher in **Mathematical Physics** with hands-on experience in **mathematical modelling**, **scientific computing** and **large-scale data and AI workflows**. Developed symbolic and numerical workflows for complex high-energy physics calculations, alongside **PyTorch** and PyTorch Geometric pipelines for **20M graph-structured samples**. Strong background in **symbolic computation**, Python-based modelling, **HPC/Slurm training**, **experiment tracking** and systematic evaluation, with interest in reliable data and AI solutions for logistics, transportation planning and optimization.

Professional Experience

University of Hamburg / DESY

Hamburg, Germany

PhD Researcher – Mathematical Modelling, Scientific Computing & AI

October 2022 – 2026

- Identified novel mathematical structures in Higgs-related particle-physics calculations and proposed ways to simplify complex analytical computations.
- Developed new mathematical building blocks for high-energy physics calculations and validated them through extensive comparison with existing research.
- Built symbolic and numerical workflows in Wolfram Mathematica to analyse complex algebraic data and test mathematical hypotheses.
- Built Python and PyTorch Geometric pipelines for more than 20 million graph-structured samples.
- Ran scalable preprocessing, feature engineering, training and evaluation workflows on HPC/Slurm infrastructure.
- Communicated results through publications, international presentations, teaching and research collaboration.

National and Kapodistrian University of Athens

Athens, Greece

Researcher - Teaching Assistant

January 2021 - September 2022

- Derived a mathematical model and analyzed its constraints and implications using differential equations, strengthening modelling and optimization-oriented problem solving.
- Supervised the Physics II laboratory course

Skills

Data Engineering & ML	Python, NumPy, Pandas, scikit-learn, preprocessing, feature engineering, structured data processing, reproducible evaluation
Data Workflows	Large-scale scientific data workflows, 20M graph samples, data/AI features, PyTorch, PyTorch Geometric, NetworkX, experiment tracking
Tools & Infrastructure	Git, Linux, Slurm, HPC workflows, W&B, MATLAB, C, quick learner of modern data-engineering tools
Scientific Computing	Wolfram Mathematica, symbolic computation, numerical analysis, algebraic modelling
Languages	English (C1), German (B2), French (B1), Greek (native), Albanian (native)

Education

University of Hamburg

PhD in Mathematical Physics

Defence: September 2026

Hamburg, Germany

October 2022 – September 2026

Utrecht University

MSc in Theoretical Physics

Utrecht, Netherlands

September 2018 – August 2020

- Final grade: 8.32/10 (Top 10%)

National and Kapodistrian University of Athens

BSc in Physics

Athens, Greece

January 2014 – November 2017

- Final grade: 8.89/10 (Top 2%)

Publications

Graph Neural Networks for the Graphical Bootstrap

arXiv:2607.03109 [hep-th], 2026.

Presented at the Physics and AI Conference, Stanford, 2026.

Novel cluster-algebraic letters for 5- and 6-point QCD processes

arXiv:2603.16743 [hep-th], 2026.

An Exceptional Cluster Algebra for Higgs + Jet Production

JHEP 01 (2025) 197.

Dynamically restoring conformal invariance in integrable σ -models

SciPost Physics 14 (2023) 043.

Awards

Hellenic Foundation for Research and Innovation

H.F.R.I. PhD Fellowship (3rd Call)

Full funding (€32,400) for a PhD project (declined in favor of a position in Hamburg).

Athens, Greece

2021

Utrecht University

Bright Minds Fellowship

Full scholarship (€20,000) awarded for outstanding academic performance.

Utrecht, Netherlands

2018